Cyrus University Course Roadmap

Tsinghua Automation, MIT EECS, Stanford Spatial AI

2026-05-15

How to use this roadmap

This PDF is the master index. A course becomes useful only when it has four outputs:

1. a source link,
2. a formula page,
3. a GoodNotes page,
4. an Obsidian/Notion evidence node.

The first individual course PDFs already wired into the web app are Stanford CS231A and CS231n. The next batch should expand MIT 6.241J, MIT 6.003, Stanford CS229, and MIT Underactuated Robotics.

One spine for all three universities

Source	Main role	Learning output
Tsinghua Automation	Automation and control engineering backbone	State, signal, feedback, estimation, robotics, intelligent systems
MIT EECS/OCW	Public engineering foundation	Algorithms, signals, dynamic systems, multivariable control, robotics tools
Stanford AI/CV	Spatial intelligence and representation learning	Camera geometry, visual representation, ML, decision-making under uncertainty

Control formula spine

State-space starts the control path:

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u}, \qquad \boldsymbol{y} = C\boldsymbol{x} + D\boldsymbol{u}.$$

Here $\boldsymbol{x}$ is state, $\boldsymbol{u}$ is input, A is the system matrix, and B is the control-input matrix.

Controllability asks whether the input can reach all state directions:

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}, \qquad \operatorname{rank}(\mathcal{C}) = n.$$

LQR turns control into an optimization problem:

$$J = \int_0^\infty \left(\boldsymbol{x}^\top Q \boldsymbol{x} + \boldsymbol{u}^\top R \boldsymbol{u} \right) dt, \qquad \boldsymbol{u} = -K\boldsymbol{x}.$$

The Riccati equation connects dynamics, cost, and feedback:

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0.$$

Estimation and spatial formula spine

Kalman filtering is the minimum estimation bridge:

$$\hat{\boldsymbol{x}}_{k|k-1} = A\hat{\boldsymbol{x}}_{k-1|k-1} + B\boldsymbol{u}_{k-1}, \qquad \hat{\boldsymbol{x}}_{k|k} = \hat{\boldsymbol{x}}_{k|k-1} + K_k(\boldsymbol{z}_k - H\hat{\boldsymbol{x}}_{k|k-1}).$$

Camera projection is the first spatial-intelligence formula:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} R & \boldsymbol{t} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}.$$

Bundle adjustment makes reconstruction an optimization problem:

$$\min_{\{T_i\},\{P_j\}} \sum_{i,j} \rho \left(\|\boldsymbol{u}_{ij} - \pi(T_i P_j)\|_2^2 \right).$$

Machine learning and decision spine

Stanford CS229 and CS231n enter through empirical risk:

$$\min_\theta \frac{1}{N} \sum_{i=1}^{N} \ell \left(f_\theta(\boldsymbol{x}_i), y_i \right) + \lambda \Omega(\theta).$$

The beginner translation is simple: choose parameters so predictions are less wrong, while regularization prevents the model from memorizing noise.

Decision-making under uncertainty enters through Bellman recursion:

$$V^\star(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^\star(s') \right].$$

This is the bridge from world models to planning: predict next states, score actions, then choose the action with the best long-term value.

Course pack

Priority	Course	Why it matters
Ready	Stanford CS231A	Camera geometry, 3D reconstruction, optical flow, estimation
Ready	Stanford CS231n	CNNs, backpropagation, visual representation, assignments
Next	MIT 6.241J	Dynamic systems, state-space control, stability
Next	MIT 6.003	Signals, systems, convolution, frequency response
Next	Stanford CS229	Core ML math for spatial models and world models
Next	MIT Underactuated Robotics	Nonlinear robotics, trajectory optimization, stability
Later	MIT 6.245	Multivariable and robust control
Later	Stanford CS224N	Language and representation side for Hermes interaction
Later	Stanford AA228/CS238	MDP, POMDP, planning under uncertainty
Expand	Tsinghua Automation	Official automation/control backbone, needs syllabus-by-syllabus expansion

GoodNotes template

For each course page, use this fixed structure:

1. Course role: one sentence.
2. Prerequisites: three nodes only.
3. Core formulas: no more than five per page.
4. One visual: block diagram, projection chain, graph, or loss landscape.
5. One output: a solved problem, a derivation, or a small interactive web scene.

Notion and Obsidian roles

Notion stores source status, PDF status, current blocker, and evidence path. Obsidian stores concept graphs and course-to-course links. GoodNotes stores handwritten derivations. The web app is the interactive front door.